

Day 11 – SQL Assignment: Comprehensive Practice

Example Table: Employee Information

Here's the SQL code to create a table for the assignment:

```
-- Drop the table if it already exists
DROP TABLE IF EXISTS employees;
-- Create the employees table
CREATE TABLE employees (
    employee_id SERIAL PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    department VARCHAR(50),
    salary DECIMAL(10, 2) CHECK (salary > 0),
    joining_date DATE NOT NULL,
    age INT CHECK (age >= 18)
);
```

Insert some sample data into the `employees` table:

```
-- Insert data into employees table
INSERT INTO employees (first_name, last_name, department, salary, joining_date, age)
VALUES
('Amit', 'Sharma', 'IT', 60000.00, '2022-05-01', 29),
('Neha', 'Patel', 'HR', 55000.00, '2021-08-15', 32),
('Ravi', 'Kumar', 'Finance', 70000.00, '2020-03-10', 35),
('Anjali', 'Verma', 'IT', 65000.00, '2019-11-22', 28),
('Suresh', 'Reddy', 'Operations', 50000.00, '2023-01-10', 26);
```

Practice Assignment Questions with Answers (Solution has given on next page)

- Q1: Retrieve all employees' first_names and their departments.
- Q2: Update the salary of all employees in the 'IT' department by increasing it by 10%.
- Q3: Delete all employees who are older than 34 years.
- Q4: Add a new column `email` to the `employees` table.
- Q5: Rename the `department` column to `dept\_name`.
- Q6: Retrieve the names of employees who joined after January 1, 2021.
- Q7: Change the data type of the `salary` column to `INTEGER`.
- Q8: List all employees with their age and salary in descending order of salary.
- Q9: Insert a new employee with the following details: 'Raj', 'Singh', 'Marketing', 60000, '2023-09-15', 30.
- Q10: Update age of employee +1 to every employee

Assignment Answers:

Q1: Retrieve all employees' first_names and their departments.

```
SELECT first_name, department FROM employees;
```

Q2: Update the salary of all employees in the 'IT' department by increasing it by 10%.

```
UPDATE employees SET salary = salary * 1.10 WHERE department = 'IT';
```

Q3: Delete all employees who are older than 34 years.

```
DELETE FROM employees WHERE age > 34;
```

Q4: Add a new column `email` to the `employees` table.

```
ALTER TABLE employees ADD COLUMN email VARCHAR(100);
```

Q5: Rename the `department` column to `dept\_name`.

```
ALTER TABLE employees RENAME COLUMN department TO dept_name;
```

Q6: Retrieve the names of employees who joined after January 1, 2021.

```
SELECT first_name, last_name FROM employees WHERE joining_date > '2021-01-01';
```

Q7: Change the data type of the `salary` column to `INTEGER`.

```
ALTER TABLE employees ALTER COLUMN salary TYPE INTEGER USING salary::INTEGER;
```

Q8: List all employees with their age and salary in descending order of salary.

```
SELECT first_name, last_name, age, salary FROM employees ORDER BY salary DESC;
```

Q9: Insert a new employee with the following details: 'Raj', 'Singh', 'Marketing', 60000, '2023-09-15', 30.

```
INSERT INTO employees (first_name, last_name, dept_name, salary, joining_date, age)  
VALUES ('Raj', 'Singh', 'Marketing', 60000, '2023-09-15', 30);
```

Q10: Update age of employee +1 to every employee

```
UPDATE employees SET age = age+1;
```